

Damien Robert

Deep Learning for Remote Sensing & Environment



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I am a postdoctoral researcher in the [EcoVision lab](#) at [University of Zurich](#), I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on "*Efficient Learning on Large-Scale 3D Point Clouds*" at [IGN](#) and [ENGIE](#), under the supervision of [Loic Landrieu](#) and [Bruno Vallet](#).

■ POSITIONS

2024–now	Postdoctoral Researcher EcoVision lab, UZH · Zurich, Switzerland Deep learning for remote sensing and environment PI: Jan Dirk Wegner
2020–2024	PhD Student LASTIG, IGN / ENGIE CRIGEN lab · Paris, France Efficient learning on large-scale 3D point clouds Advisors: Loic Landrieu , Bruno Vallet
2017–2020	R&D Engineer SIRADEL / ENGIE · Rennes, France Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data
2017	Co-Founder Inspirama · Rennes, France Website gathering book recommendations from inspiring people
2015	R&D Intern Dassault · Providence, RI, USA Dimensionality reduction and dynamic system modeling
2014	R&D Intern Dassault · Providence, RI, USA UX design

■ EDUCATION

2011–2015	MSc, Engineering École Centrale de Lyon · Lyon, France Mathematics, Computer Science, Mechanics, Signal Processing, Automation
2022	Summer School ICVSS · Sicily, Italy Computer vision courses by world-renowned experts in academia and industry
2017	AI Fall School CNRS · Lyon, France Multi-disciplinary course for AI students and researchers

■ AWARDS

2026	Outstanding Reviewer · CVPR
2025	PhD Award accessit · AFRIF French Association for Pattern Recognition and Interpretation
2024	Outstanding Reviewer · ECCV
2022	Best Paper Finalist · CVPR Top 0.4% of submissions

2026

Toward a Foundation Model for Forest Point Clouds

Y. Yue, S. Puliti, **D. Robert**, A. Topaloğlu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler
Under review

From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps

G. Sialelli, R. Young, Y. Jiang, C. Aybar, L. Scheibenreif, **D. Robert**, C. Mosig, M. Czerkawski, A. J. Stewart, J. D. Wegner,
 A. Pirinen, O. Mogren, K. Schindler
Under review

Bias Aware Benchmark for Species Distribution Modeling

E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, **D. Robert**, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia
[World Biodiversity Forum](#)

LitePT: Lighter Yet Stronger Point Transformer

Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler
[CVPR](#) [Highlight](#)

EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation

L. Geist, L. Landrieu, **D. Robert**
[ICRA](#)

FORMSPoT: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series

M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciaïa

[Remote Sensing of Environment](#)

2025

MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation

K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner
[PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science](#)

SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology

E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann
[CVPR workshop EarthVision](#)

GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance

K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner
[ISPRS Geospatial Week](#) [Oral](#)

Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks

J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner
[ICLR workshop Tackling Climate Change with Machine Learning](#)

Lossy Neural Compression for Geospatial Analytics: A Review

C. Gomes, I. Wittmann, **D. Robert**, J. Jakubik, T. Reichelt, M. Martone, S. Maurogiovanni, R. Vinge, J. Hurst, E. Scheurer,
 R. Sedona, T. Brunschweiler, S. Kesselheim, M. Batic, P. Stier, J. D. Wegner, G. Cavallaro, E. Pebesma, M. Marszałek, M.
 A. B. Plomer, K. Adriko, P. Fraccaro, R. Kienzler, R. Briq, S. Benassou, M. Lazzarini, C. M. Albrecht
[IEEE Geoscience and Remote Sensing Magazine](#)

2024

Efficient Learning on Large-Scale 3D Point Clouds

D. Robert
[PhD Thesis, Université Gustave Eiffel](#) [AFRIF PhD award](#)

Jury: Sébastien Lefèvre, Cédric Démonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loïc Landrieu, Bruno Vallet

Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering

D. Robert, H. Raguet, L. Landrieu
[3DV](#) [Oral](#) · top 5.3% of submissions

2023

Efficient 3D Semantic Segmentation with Superpoint Transformer

D. Robert, H. Raguet, L. Landrieu
[ICCV](#) · top 26.8% of submissions

2022

Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation

D. Robert, B. Vallet, L. Landrieu
[CVPR](#) [Best paper finalist](#) [Oral](#) · top 0.4% of submissions

■ COLLABORATION & SUPERVISION

Collaborators	Konrad Schindler ETH · Christian Rupprecht Oxford · Jan Dirk Wegner UZH · Loic Landrieu ENPC · Vivien Sainte Fare Garnot UZH · Lukas Drees UZH · Martin Schwartz LSCE · Elena Plekhanova WSL
PhD Students	Emilia Arens UZH with Jan Dirk Wegner · Johannes Dollinger UZH with Jan Dirk Wegner · Kaan Karaman UZH with Jan Dirk Wegner · Louis Geist ENPC with Loic Landrieu · Yuanwen Yue ETH with Konrad Schindler and Christian Rupprecht
Master Students	Valerio Schelbert WSL · Jackson Sunny École Polytechnique with SAMP
Advising	UpCircle AI Switzerland · École des Sciences Criminelles de Lausanne Switzerland · SAMP AI France · GEODIGITAL US

■ REVIEWING, CHAIRING & ORGANISING

 Reviewer  Area chair  Organising committee

2027	 CVPR workshop EarthVision
2026	 ECCV ·  CVPR
2025	 CVPR ·  CVPR workshop EarthVision
2024	 ECCV ·  ISPRS Journal of Photogrammetry and Remote Sensing ·  ICLR workshop ML4RS ·  CVPR workshop EarthVision
2023	 CVPR ·  CVPR workshop EarthVision
2022	 ISPRS Journal of Photogrammetry and Remote Sensing ·  CVPR workshop EarthVision

■ TEACHING

2026	Transformer · UZH · Lecture, Master 2
2025	AI4Good group projects · UZH · Labs, Master 2
2025	Transformer · UZH · Lecture, Master 2
2024	Transformer, NeRFs, and Diffusion · UZH · Lecture and labs, Master 2
2023	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2
2022	3D Deep Learning for Remote Sensing · ISPRS'22 · Tutorial, Researchers
2022	3D Deep Learning, Torch-Points3D & DeepViewAgg · ENGIE CRIGEN lab · Tutorial, Researchers
2022	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2
2020	Deep Learning for Computer Vision · Ecole Polytechnique · Lecture, Master 1

■ OPEN-SOURCE SOFTWARE

★ 406  45	LitePT ·  prs-eth/LitePT
★ 15  0	torch-graph-components ·  drprojects/torch-graph-components
★ 1055  136	Superpoint Transformer ·  drprojects/superpoint_transformer
★ 53  3	NoRA ·  drprojects/nora
★ 74  8	Point Geometric Features ·  drprojects/point_geometric_features
★ 237  24	DeepViewAgg ·  drprojects/DeepViewAgg

■ SELECTED TALKS

2026	ESA-NASA workshop on AI Foundation Model for EO · tutorial · Building Earth Observation Embedding Workflows with TerraTorch · Virtual ESA Φ-lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy
2025	MPI of Animal Behavior · Research at EcoVision lab · Zurich, Switzerland Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual NatureFinance · Research at EcoVision lab · Virtual NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual
2024	3D Data Academy · interview · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube Google DeepMind · Research at EcoVision lab · Zurich, Switzerland 3DV'24 · oral · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland
2023	Ecole des Ponts, IMAGINE lab · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France
2022	CV News · interview · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US CVPR'22 · oral · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US
45 talks given since 2021 — full list at drprojects.github.io/talks .	

■ SKILLS & LANGUAGES

Research	3D & geometry · Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning			
	Efficient & scalable learning · Efficient deep learning · Scalable model design · EO data representation and compression			
	Learning paradigms · Multimodal learning · Self-supervised learning · Representation learning			
	Environment & Earth observation · Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology			
Tools	 Python	 PyTorch	 PyTorch Lightning	 PyTorch Geometric
	 C++	 Hydra	 Weights & Biases	 Git
	 Linux	SLURM	 scikit-learn	 Plotly
	 Blender	LaTeX		
Languages	French 	Native	English 	Fluent
	Spanish 	Intermediate	German 	Beginner
Abroad	Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015			
Interests	Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games			